1 Model

1.1 Model Set-Up

Agents. There is a unit continuum of domestic suppliers, indexed by $i \in [0,1]$, a finite set $\mathcal{J} = \{1, \ldots, J\}$ of domestic buyers, and a finite set $\mathcal{C} = \{1, \ldots, C\}$ of foreign destinations. A subset $\mathcal{J}^M \subseteq \mathcal{J}$ consists of domestic affiliates of multinational corporations (MNCs); buyers in $\mathcal{J} \setminus \mathcal{J}^M$ are purely domestic firms. Suppliers produce differentiated varieties and may sell them to domestic buyers or export them to foreign destinations. Domestic buyers combine supplier varieties with labor in production. The domestic production side of the economy is therefore represented by a bipartite network with linkages between suppliers and buyers.

Buyers. A buyer j is characterized by its productivity $\alpha_j > 0$. It combines a CES composite of its suppliers' varieties with labor L_j to produce a differentiated final good. The composite and output are

$$Q_j = \left[\int_{i:j\in S_i} m_{ij}^{1/\sigma} x_{ij}^{(\sigma-1)/\sigma}\, di\right]^{\sigma/(\sigma-1)}, \tag{1}$$

$$Y_j = \alpha_j \left[\omega Q_j^{(\xi-1)/\xi} + (1-\omega) L_j^{(\xi-1)/\xi}\right]^{\xi/(\xi-1)}, \tag{2}$$

where $S_j = \{i : j \in S_i\}$ is the set of suppliers that have activated a link with buyer j, $\sigma > 1$ is the elasticity of substitution across supplier varieties, $\xi > 0$ the elasticity between the supplier composite and labor, and $\omega \in (0,1)$. The weight $m_{ij} \geq 0$ is a match-specific demand shifter: matches differ in the breadth of the product scope the buyer sources from the supplier. We take m_{ij} to be log-normal with unit mean, independent across pairs and of all other primitives. The price index of the composite is

$$P_j = \left[\int_{i:j\in S_i} m_{ij}\, p_i^{1-\sigma}\, di\right]^{1/(1-\sigma)}, \tag{3}$$

and cost minimization with constant-markup pricing gives the buyer's marginal cost and final-good price,

$$MC_j = \frac{1}{\alpha_j}\left[\omega^\xi P_j^{1-\xi} + (1-\omega)^\xi w^{1-\xi}\right]^{1/(1-\xi)}, \qquad p_j^Y = \frac{\gamma}{\gamma-1} MC_j. \tag{4}$$

Representative household. A representative household supplies labor inelastically, $\bar{L}$, and allocates its expenditure across domestic final goods and an imported foreign good in a two-tier nest. The upper tier is a CES aggregator with Armington elasticity $\rho > 0$ over a domestic final-good composite Q^H and the imported good Q^M, placing weight $\nu \in (0,1)$ on the domestic composite; the lower tier is a CES aggregator with elasticity $\gamma > 1$ over the individual buyers' final goods. The imported good is supplied perfectly elastically by the rest of the world and serves as the *numeraire*, with its price normalized to one, so the wage w is the economy's real exchange rate. Writing $P^H = \left[\sum_{k\in\mathcal{J}} (p_k^Y)^{1-\gamma}\right]^{1/(1-\gamma)}$ for the price index of the domestic composite and $P^U = \left[\nu (P^H)^{1-\rho} + (1-\nu)\right]^{1/(1-\rho)}$ for the consumer price index, household expenditure E^H splits between the domestic composite and imports as

$$E^{H,H} = \nu\left(\tfrac{P^H}{P^U}\right)^{1-\rho} E^H, \qquad M = (1-\nu)\,(P^U)^{\rho-1} E^H, \qquad E^{H,H} + M = E^H. \tag{5}$$

Within the domestic composite, buyer j faces the residual demand

$$Y_j = \Theta (p_j^Y)^{-\gamma}, \qquad \Theta = \frac{E^{H,H}}{\sum_{k\in\mathcal{J}} (p_k^Y)^{1-\gamma}}, \tag{6}$$

where Θ is the common domestic demand level, scaled by the share $E^{H,H}$ of expenditure that falls on domestic goods. The expenditure level E^H and the wage w are pinned down in general equilibrium (Section 1.3). A higher α_j lowers buyer j's marginal cost and raises its scale, generating a larger demand shifter B_j for its suppliers.

Multinational corporations. A multinational corporation is a parent firm p with productivity α_p. The parent can operate affiliates across locations, including the domestic economy, by paying location-specific fixed costs. In particular, a parent operating in location ℓ earns operating profit proportional to $\alpha_p^{\sigma-1} M_\ell$, where M_ℓ summarizes market size, and pays a fixed cost $F_{p\ell}$. It enters location ℓ whenever

$$\alpha_p^{\sigma-1} M_\ell \geq F_{p\ell}. \tag{7}$$

The problem is separable across locations, so the multinational network of affiliates is a collection of independent threshold decisions rather than a combinatorial choice problem. The set $\mathcal{J}^M$ of domestic MNC affiliates and each parent's foreign footprint are the realized outcomes of these entry decisions. A domestic affiliate $m \in \mathcal{J}^M$ inherits its parent's productivity, $\alpha_m = \alpha_p$, so selection into the domestic market implies that MNC buyers are more productive on average than purely domestic buyers. If entry costs $F_{p\ell}$ are independent of parent productivity, more productive parents are also more likely to clear the entry threshold in any given location and therefore have broader foreign footprints in expectation. We denote the foreign footprint of domestic MNC buyer m by $g_{mc} = \mathbf{1}\{\text{the parent of } m \text{ operates an affiliate in destination } c\}$ and define $K_c = \{m \in \mathcal{J}^M : g_{mc} = 1\}$ as the set of domestic MNC buyers whose parent is present in destination c (time-indexed K_{ct} in the panel). For non-MNC firms $j \notin \mathcal{J}^M$, set $g_{jc} = 0$.

Suppliers. Each supplier i produces a single horizontally differentiated variety using labor only. Its production technology is

$$y_i = \theta_i(S)\ell_i, \tag{8}$$

where ℓ_i is labor hired competitively at wage w, and $\theta_i(S)$ is supplier productivity, which depends on the set S of active buyer relationships. A supplier can sell its variety to domestic buyers with which it has activated a relationship and can also export to foreign destinations after paying destination-specific export costs. Domestic demand is generated by the buyers in S, while foreign demand is generated by destination markets $c \in \mathcal{C}$.

Productivity spillovers from buyer linkages. Active buyer relationships raise supplier productivity. A supplier i is endowed with a fundamental productivity component $\bar{\theta}_i > 0$. Each buyer j belongs to one of a finite set of categories $\mathcal{K}$, with category label $k(j) \in \mathcal{K}$ (for instance $\mathcal{K} = \{\textsc{dom}, \textsc{mne}\}$), and the supplier carries a category-specific absorptive capacity. The learning technology is log-additive in active buyer relationships, with category-specific per-link spillovers:

$$\log \theta_i(S) = \log \bar{\theta}_i + \sum_{j \in S} \beta_i^{k(j)} \alpha_j, \qquad \beta_i^k \geq 0, \tag{9}$$

where β_i^k is the supplier's category-k absorptive-capacity slope; we collect these in the absorptive-capacity type $\beta_i \equiv (\beta_i^k)_{k \in \mathcal{K}}$. For compactness we write $\psi_{ij} \equiv \beta_i^{k(j)} \alpha_j \geq 0$ for the per-link log-spillover. Thus, absent active buyer relationships, $\theta_i(\varnothing) = \bar{\theta}_i$; adding buyer j raises supplier i's productivity by the proportional amount $\exp(\psi_{ij}) - 1$, increasing in buyer productivity α_j and in the supplier's absorptive capacity. This specification captures the idea that more productive buyers transmit more valuable knowledge, quality requirements, and managerial practices to their suppliers, while suppliers differ in their ability to absorb these improvements. Because the spillover differs by category, a link to an MNC buyer can transmit a larger productivity gain than a link to a domestic buyer of equal productivity—through $\beta_i^{\text{MNE}} \geq \beta_i^{\text{DOM}}$—over and above the difference already implied by selection on α_j.

Matching. Suppliers can only activate a relationship with a buyer it has previously met. Meetings with buyer j arrive according to an independent Poisson process with rate $\lambda_j(\bar{\theta}_i/\bar{\theta}_{\text{med}})^{\varkappa}$, where $\lambda_j > 0$, $\varkappa \geq 0$, and $\bar{\theta}_{\text{med}}$ is the median of the baseline-productivity distribution: larger suppliers maintain more commercial relationships and therefore meet more buyers, with $\varkappa = 0$ recovering the size-independent case. Equivalently, over the relevant period the probability that supplier i meets buyer j is

$$\Pr[j \in M_i] = 1 - e^{-\lambda_j(\bar{\theta}_i/\bar{\theta}_{\text{med}})^{\varkappa}}. \tag{10}$$

Meetings are independent across buyers and across suppliers, identically distributed across suppliers only when $\varkappa = 0$, so the met set $M_i \subseteq \mathcal{J}$ is a random draw from these Poisson meeting processes. After observing M_i, the supplier chooses its active set $S \subseteq M_i$.[1]

Foreign markets. In addition to serving domestic buyers, suppliers may export to foreign destinations $c \in \mathcal{C}$. Reaching destination c entails a bilateral iceberg trade cost $\tau_{ic} \geq 1$: delivering one unit requires shipping τ_{ic} units, so the supplier's delivered price is $\tau_{ic}p_i$. Demand in each destination is isoelastic, summarized by a destination-specific demand shifter $D_c > 0$ and elasticity of substitution $\sigma > 1$, so that export demand for the supplier's variety is $x_{ic}^X = (\tau_{ic}p_i)^{-\sigma}D_c$. For convenience we define the effective firm-destination demand shifter as $\tilde{D}_{ic} \equiv \tau_{ic}^{1-\sigma}D_c$. Exporting to c also requires paying a destination-specific fixed cost $f_c^X(S)$, which may depend on the supplier's active MNC relationships, as specified next.

Market-access spillovers from MNC linkages. A link to a domestic MNC affiliate whose parent is present in destination c may provide commercial intelligence, reputation, or referrals that ease entry into that market. We model this as an additive reduction in the destination-specific export fixed cost:

$$f_c^X(S) = \bar{f}_c - \sum_{m \in S \cap K_c} \eta_{mc}, \qquad \eta_{mc} \geq 0, \tag{11}$$

where $\bar{f}_c$ is the baseline fixed cost of exporting to c and η_{mc} is the reduction generated by a link to MNC buyer m present in destination c, $g_{mc} = 1$. We impose the parameter restriction $\bar{f}_c - \sum_{m \in A \cap K_c} \eta_{mc} \geq \underline{f}_c > 0$ for every feasible active set A.[2] This channel differs from productivity spillovers in two ways: (i) it operates only through MNC links and, (ii) only in destinations reached by those MNCs' network of affiliates. It is therefore destination-specific rather than a general channel affecting firm performance.

[1] Note that meeting a buyer does not by itself generate spillovers: only an activated link to buyer j contributes to productivity through α_j, and only an activated link to an MNC buyer $m \in \mathcal{J}^M$ can reduce export fixed costs in destinations c with $g_{mc} = 1$.

[2] The lower bound is imposed as a restriction on admissible parameter values, not as a truncation of the fixed-cost function. That is, we restrict attention to parameter vectors such that $\bar{f}_c - \sum_{m \in S \cap K_c} \eta_{mc} \geq \underline{f}_c > 0$ for every feasible active set S. We do not replace $f_c^X(S)$ with $\max\{\underline{f}_c, \bar{f}_c - \sum_{m \in S \cap K_c} \eta_{mc}\}$. This distinction matters because the baseline access technology remains additive in active MNC links, which preserves the lattice structure used in the supplier problem.

Timing. The model is static within a period, and we suppress the time index except in the empirical specifications, where t indexes the panel. Within a period, supplier i enters with its type $(\bar{\theta}_i, \beta_i, \varphi_i)$, where φ_i is the link-cost shifter introduced in Section 1.2; the meeting set M_{it} is realized, together with the match shifters m_{ij} of the buyers met; the supplier chooses its active links $S_{it} \subseteq M_{it}$, which determine productivity $\theta_i(S_{it})$ and export fixed costs $f^X_{ct}(S_{it})$; it then chooses export destinations; and production, pricing, and trade occur.

1.2 Supplier Link Choice and Supermodularity

Operating profits. Supplier i has productivity $\theta_i(S)$ and constant marginal cost $w/\theta_i(S)$. Facing the CES demand of any buyer j it supplies,

$$x_{ij} = m_{ij} \left(\frac{p_i}{P_j} \right)^{-\sigma} Q_j, \tag{12}$$

with the input price index P_j given by (3), the supplier sets the price implied by the constant CES markup,

$$p_i(S) = \frac{\sigma}{\sigma - 1} \frac{w}{\theta_i(S)}. \tag{13}$$

The price is identical across buyers because marginal cost is common across markets. Revenue from buyer j is $r_{ij} = m_{ij}\, p_i(S)^{1-\sigma} P_j^{\sigma-1} E_j$, where $E_j = P_j Q_j$ is buyer j's expenditure on supplier varieties, and operating profit is the constant share $\pi_{ij} = r_{ij}/\sigma$. Writing $\kappa \equiv (\frac{\sigma}{\sigma-1} w)^{1-\sigma}$ and $B_j \equiv P_j^{\sigma-1} E_j$,

$$\pi_{ij}(S) = \frac{\kappa}{\sigma}\, m_{ij}\, \theta_i(S)^{\sigma-1} B_j. \tag{14}$$

The same algebra applies to destination c, with the iceberg cost folded into the effective demand shifter $\widetilde{D}_{ic} = \tau_{ic}^{1-\sigma} D_c$, which is taken as given by the supplier. Export operating profit is thus

$$\pi^X_{ic}(S) = \frac{\kappa}{\sigma} \theta_i(S)^{\sigma-1} \widetilde{D}_{ic}. \tag{15}$$

The shifters B_j thus collect everything about the buyer j's demand that is common across its suppliers, while m_{ij} carries the match-specific component; the supplier takes both as given. B_j is increasing in buyer productivity α_j and is endogenized in general equilibrium in Section 1.3.

The supplier's problem. Activating a link j requires a fixed cost $\varphi_i f_j$, with $f_j > 0$, borne by the supplier; the shifter $\varphi_i > 0$ is supplier-specific, log-normal with unit mean, i.i.d. across suppliers, and drawn with the supplier's type: suppliers differ in how costly relationship formation is.[3] We abstract from buyer-side screening and relationship costs, keeping link formation one-sided: conditional on a meeting, the supplier activates the link by paying $w\varphi_i f_j$, and the buyer then demands its variety at the CES price. Given its met set M_i, supplier i chooses an active buyer set $S \subseteq M_i$ and an export destinations set $X \subseteq \mathcal{C}$ to maximize

$$\Pi_i(S, X) = \frac{\kappa}{\sigma} \theta_i(S)^{\sigma-1} \sum_{j \in S} m_{ij} B_j + \sum_{c \in X} \left[\frac{\kappa}{\sigma} \theta_i(S)^{\sigma-1} \widetilde{D}_{ic} - w f^X_c(S) \right] - w\varphi_i \sum_{j \in S} f_j. \tag{16}$$

For any active buyer set S, the optimal export set is

$$X_i(S) = \left\{ c \in \mathcal{C} : \frac{\kappa}{\sigma} \theta_i(S)^{\sigma-1} \widetilde{D}_{ic} \geq w f^X_c(S) \right\}. \tag{17}$$

[3]Fixed costs are denominated in units of labor: activating link j costs $w\varphi_i f_j$ and exporting to c costs $w f^X_c(S)$, where w is the wage determined in general equilibrium (Section 1.3).

Substituting this optimal export decision into the objective gives the suppliers' link-choice problem

$$\Pi_i(S) = \frac{\kappa}{\sigma}\theta_i(S)^{\sigma-1}\sum_{j\in S} m_{ij}B_j + \sum_{c\in\mathcal{C}}\left[\frac{\kappa}{\sigma}\theta_i(S)^{\sigma-1}\widetilde{D}_{ic} - wf_c^X(S)\right]^+ - w\varphi_i\sum_{j\in S} f_j, \qquad S\subseteq M_i. \quad (18)$$

The first term is domestic operating profit over active buyers. The supplier's productivity $\theta_i(S)$ depends on the whole active set through the productivity spillover technology, so adding a link affects not only sales to the new buyer but also the profitability of all existing domestic and foreign markets. The second term is export profit net of destination fixed costs, with $[\cdot]^+$ recording the optimal export-entry decision destination by destination. The third term is the cost of active domestic links. Because $f_c^X(S)$ falls with the supplier's active MNC links whose parents are present in c, the link choice and export-entry decisions are coupled.

Combinatorial discrete choice and single crossing. The supplier's link problem is combinatorial because the payoff from an active relationship depends on the rest of the active set. Productivity $\theta_i(S)$ is determined by all active buyer links, while export fixed costs $f_c^X(S)$ depend on the subset of active MNC links whose parents are present in destination c. Hence the supplier cannot evaluate each link independently: it must choose among the $2^{|M_i|}$ subsets of its met set. For $S\subseteq M_i$ and $j,k\notin S$, we define the marginal value of link j and the cross-difference between links j and k as

$$\delta_j\Pi_i(S) = \Pi_i(S\cup\{j\}) - \Pi_i(S) \quad (19)$$

and

$$D_{jk}\Pi_i(S) = \Pi_i(S\cup\{j,k\}) - \Pi_i(S\cup\{j\}) - \Pi_i(S\cup\{k\}) + \Pi_i(S). \quad (20)$$

A positive cross-difference means that the two links are complements at S; the next paragraph establishes that $D_{jk}\Pi_i(S)\geq 0$ throughout, which is what delivers the single-crossing property we use here. That property is single-crossing differences in choices from below: for all $S\subseteq S'\subseteq M_i\setminus\{j\}$,

$$\delta_j\Pi_i(S)\geq 0 \quad\Longrightarrow\quad \delta_j\Pi_i(S')\geq 0. \quad (21)$$

This condition rules out reversals in the sign of a link's marginal value as the active set expands. It is the monotonicity condition underlying the algorithm for combinatorial discrete choice in Arkolakis, Eckert, and Shi (2026), which builds on the monotonicity insight in Jia (2008) for high-dimensional entry problems. Rather than enumerating all feasible subsets, the algorithm uses this monotonicity to eliminate regions of the choice lattice and solve the supplier link choice problem globally.

Supermodularity. The baseline parameterization delivers this single-crossing property through a stronger condition: supermodularity of the supplier objective. The economic mechanism is that buyer linkages are complementary through the productivity spillover. A link to one buyer raises supplier productivity proportionally, and because CES operating profits scale with $\theta_i(S)^{\sigma-1}$, this productivity improvement raises the value of serving other buyers and foreign destinations. This complementarity rests on two features of the learning technology, and on neither the levels nor the cross-category dispersion of the spillovers: the per-link contributions are additively separable in logs, so $\Phi(S)=\theta_i(S)^{\sigma-1}$ is the exponential of a modular function; and each per-link increment is nonnegative. The per-link spillover may therefore differ arbitrarily across buyer categories—larger for MNC than for domestic buyers—without affecting supermodularity, which is a property of the interaction between links rather than of their marginal levels. The market-access channel preserves this lattice structure in the baseline because fixed-cost reductions are additive across MNC links; each active MNC relationship lowers the relevant destination fixed cost by its own amount, independently of the other active links. In other words, the market-access channel is modular. Link fixed costs are additive as well. The match shifters m_{ij} and the cost shifter φ_i enter the supplier's problem only as positive constants, rescaling the demand weights to $m_{ij}B_j$ and the link costs to $\varphi_i f_j$, so they leave supermodularity unaffected. Marginal link values are thus increasing in the active set, so the supplier objective satisfies SCD-C from below—the property the squeezing algorithm of Arkolakis, Eckert, and Shi (2026) requires. Appendix A.1 proves this formally.

Supermodularity also delivers comparative statics for optimal link sets. Let $\overline{S}_i$ denote the greatest optimal link set. Holding fixed the other components of the supplier's type, Topkis's theorem (Topkis, 1998) implies that $\overline{S}_i$ expands when supplier productivity $\bar{\theta}_i$, any absorptive-capacity slope β_i^k, buyer demand shifters B_j, or destination demand shifters D_c increase, and contracts when link costs f_j or baseline export fixed costs $\bar{f}_c$ increase. Appendix A.2 provides the formal proof. The appendix also shows that, for fixed $(\beta_i, \varphi_i, \{m_{ij}\})$ and under a generic no-ties condition, the optimal link policy is a step function of supplier productivity. As $\bar{\theta}_i$ rises, the supplier changes its active set only at finitely many indifference thresholds. This threshold structure is used for exact aggregation in Section 1.3.

1.3 General Equilibrium

Demand shifters. General equilibrium closes the model by endogenizing the domestic demand shifters $\{B_j\}$ that suppliers take as given in their link-choice problem. The foreign demand shifters $\{D_c\}$ remain exogenous, reflecting a small-economy assumption for export markets; symmetrically, the economy imports a foreign good at the exogenous world price (the numeraire), and the wage w and expenditure level E^H adjust to clear the labor market and the household budget.

We treat suppliers as a continuum of atomistic firms. The role of this assumption is to make aggregation transparent. Individual suppliers do not internalize their effect on buyer-level objects, but the distribution of supplier types and link choices determines each buyer's input price index P_j. Given P_j, buyer j's demand shifter is

$$B_j = P_j^{\sigma-1} E_j, \tag{22}$$

where E_j is expenditure on the supplier composite. Because supplier policies are step functions of baseline productivity $\bar{\theta}_i$, these aggregates can be computed exactly by integrating over the supplier productivity distribution. Appendix A.3 provides the aggregation formulas and derivations.

Competition among Suppliers. The main general-equilibrium force is competition among suppliers serving the same buyer. When more suppliers serve buyer j, the buyer's input price index P_j falls. This benefits the buyer, but it also reduces the residual demand faced by each incumbent supplier. The strength of this response is governed by

$$\frac{\partial \ln B_j}{\partial \ln P_j} = (\sigma - \xi) + (\xi - \gamma) s_j^M, \tag{23}$$

where $s_j^M \in [0, 1]$ is buyer j's intermediate-input cost share. Since the right-hand side is affine in s_j^M, it is positive for every buyer if and only if $\sigma > \max\{\xi, \gamma\}$. Under that condition, a fall in P_j lowers B_j: as the supplier base expands, incumbent suppliers face lower demand. This is a demand-side general-equilibrium condition; the supplier's link-choice problem is supermodular under $\sigma > 1$ and $\beta_i \geq 0$ alone, independent of ξ and γ (Appendix A.1).

Equilibrium. An equilibrium is a collection of allocations and prices

$$\mathcal{E} = \Big(w, \Theta, E^H, M, \{B_j, P_j, E_j, Q_j, Y_j, p_j^Y, \ell_j^B\}_{j\in\mathcal{J}}, \{S_i, X_i, p_i, y_i, \ell_i\}_{i\in[0,1]}\Big)$$

such that buyers optimize according to (4), suppliers price according to (13), suppliers choose linkages and exports according to (16), and the following markets clear:

1. *Final-good market clearing.* Each buyer's output equals the household demand in (6).

2. *Intermediate-good market clearing.* Each supplier's output equals demand from its active buyers and export destinations,

$$y_i = \theta_i(S_i)\ell_i = \sum_{j\in S_i} x_{ij} + \sum_{c\in X_i} \tau_{ic} x_{ic}^X; \qquad x_{ij} = m_{ij}\, p_i^{-\sigma} B_j, \quad x_{ic}^X = (\tau_{ic} p_i)^{-\sigma} D_c. \tag{24}$$

3. *Household budget constraint.* Total household expenditure on domestic goods and imports equals labor income plus the profits of domestically owned firms,

$$\underbrace{\sum_{j\in\mathcal{J}} p_j^Y Y_j}_{E^{H,H}} + M = w\bar{L} + \int_0^1 \Pi_i(S_i, X_i)\, di + \sum_{j\in\mathcal{J}\setminus\mathcal{J}^M} \Pi_j^B; \qquad \Pi_j^B = p_j^Y Y_j - E_j - w\ell_j^B, \tag{25}$$

   where the profits of MNC affiliates $m \in \mathcal{J}^M$ are repatriated to their foreign parents and thus do not enter domestic income.[4]

4. *Labor-market clearing.* Labor demand from supplier production, buyer production, and fixed costs equals the endowment,

$$\bar{L} = \int_0^1 \ell_i\, di + \sum_{j\in\mathcal{J}} \ell_j^B + \int_0^1 \left(\varphi_i \sum_{j\in S_i} f_j + \sum_{c\in X_i} f_c^X(S_i)\right) di. \tag{26}$$

For computation, the equilibrium can be summarized by the fixed point $B^* = \Psi(B^*)$ where $\Psi_j(B) = P_j(B)^{\sigma-1} E_j(B)$. Appendix A.3 constructs this operator, gives the aggregation formulas, proves its existence and discusses multiplicity.

[4] The imported good is the numeraire, so the wage w is the real exchange rate, pinned by labor-market clearing: a higher w raises domestic costs relative to the fixed foreign demand $\{D_c\}$ and contracts exports, so labor demand is decreasing in w and clears for some $w > 0$ at any endowment. The household budget then determines the expenditure level E^H, and hence the domestic scale Θ through (5). Imports close the trade account: by Walras' law, at the labor-clearing wage the balance of payments holds,

$$\underbrace{\int_0^1 \sum_{c\in X_i} \sigma\, \pi_{ic}^X(S_i)\, di}_{\text{export revenue}} = M + \underbrace{\sum_{m\in\mathcal{J}^M} \Pi_m^B}_{\text{repatriated profit}},$$

so exports are *not* tied to repatriation. Absent MNC affiliates the right-hand side is just M: exports are financed by imports rather than by repatriated MNE profits.

1.4 Identification and Empirical Content

Export entry in the model. The model implies a discrete export-entry decision at the firm–destination level. By (17), supplier i exports to destination c in period t if and only if export operating profit covers the destination fixed cost:

$$\text{Export}_{ict} = \mathbf{1}\{y^*_{ict} \geq 0\}, \qquad y^*_{ict} = \log\left(\frac{\kappa_t}{\sigma}\,\theta_i(S_{it})^{\sigma-1}\tau_{ic}^{1-\sigma}D_{ct}\right) - \log\left(w_t f^X_{ct}(S_{it})\right). \tag{27}$$

Taking logs and substituting the learning technology (9) and the access technology (11) makes both spillover channels explicit in the latent entry index:

$$y^*_{ict} = \underbrace{(\sigma-1)\log\bar{\theta}_i}_{\text{firm productivity}} + \underbrace{(\sigma-1)\sum_{j\in S_{it}}\psi_{ij}}_{\text{productivity spillovers}} + \underbrace{\mu_{ct}}_{\text{destination-year}} + \underbrace{\chi_{ic}}_{\text{bilateral}} + \underbrace{a_{ict}}_{\text{market access spillovers}}, \tag{28}$$

where $\psi_{ij} = \beta_i^{k(j)}\alpha_j$ is the per-link log-spillover of (9), $\mu_{ct} \equiv \log D_{ct} - \log\bar{f}_{ct} + \log(\kappa_t/(\sigma w_t))$ collects destination-year demand and baseline export costs together with macro terms common to all destinations, $\chi_{ic} \equiv (1-\sigma)\log\tau_{ic}$ captures persistent bilateral trade costs, and

$$a_{ict} \equiv \log\bar{f}_{ct} - \log f^X_{ct}(S_{it}) = -\log\left(1 - \tfrac{1}{\bar{f}_{ct}}\textstyle\sum_{m\in S_{it}\cap K_{ct}}\eta_{mc}\right) \;\geq\; 0 \tag{29}$$

is the market-access discount generated by active MNC linkages, strictly positive only if the supplier has an active link to at least one MNC whose parent is present in destination c ($S_{it}\cap K_{ct} \neq \varnothing$), where the footprint sets K_{ct} carry a time index because parents open affiliates in new destinations over the sample period.

The decomposition separates the two channels by their incidence across destinations. The productivity spillover term is *destination-neutral*: it changes only when the active set S_{it} changes, and when it does it shifts the entry index by the same amount in every destination c. The market-access discount is *destination-specific*: within the same firm–year it takes different values across destinations, according to whether the supplier's MNC buyers have a parent presence there. Every other component is either fixed at the firm level ($\bar{\theta}_i$), bilateral and persistent (χ_{ic}), or common to all firms in a destination–year (μ_{ct}). This incidence structure is what the empirical design exploits: a single firm–destination–year regression can carry *both* channels as separate regressors, with firm-destination and destination-year fixed effects absorbing the remaining components.

Empirical specification. (28) thus maps into a single firm-destination-year specification, which we take to the data in Section **??**:

$$\text{Export}_{ict} = \zeta_\theta\,\text{Linked}_{i,t-1} + \zeta_a\,(\text{Linked}\times\text{Present})_{ic,t-1} + \delta_{ic} + \mu_{ct} + u_{ict}, \tag{30}$$

where $\text{Linked}_{i,t-1} = \mathbf{1}\{S_{i,t-1}\cap\mathcal{J}^M \neq \varnothing\}$ indicates that the supplier sells to at least one MNC buyer and is the extensive-margin counterpart of the productivity spillover term in (28). It is destination-neutral, so its coefficient ζ_θ measures the shift in export participation that an MNC linkage generates in *every* destination. $(\text{Linked}\times\text{Present})_{ic,t-1} = \mathbf{1}\{S_{i,t-1}\cap K_{c,t-1} \neq \varnothing\}$ is a binary variable that takes value one if the supplier sells to at least one MNC whose parent operates an affiliate in destination c, and is the extensive-margin counterpart of the market-access discount a_{ict}. It is destination-specific, so its coefficient ζ_a measures the additional probability of exporting to the destinations reached by the linked MNCs' affiliate networks. The firm–destination fixed effect δ_{ic} absorbs the firm productivity component $(\sigma-1)\log\bar{\theta}_i$ and the bilateral cost $(1-\sigma)\log\tau_{ic}$;

the destination–year fixed effect absorbs μ_{ct}. We estimate (30) as a linear probability model on the full analysis sample and, separately, for firms with no prior exports to destination c, where the outcome records the first crossing of the entry threshold in (27). All specifications control for the forward-linkage counterpart of the access treatment—buying from an MNC present in c—and firm size. In the main panel implementation, the MNC network is measured using the 2017 D&B snapshot; the event study separately exploits the timing of new affiliate openings. The coefficients ζ_θ and ζ_a are reduced-form counterparts of the model's productivity and market-access terms, rather than direct estimates of ψ_{im} and η_{mc}.

Identifying market-access spillovers. ζ_a is identified from variation across destinations within a firm: whether a supplier is more likely to export to the destinations where its MNC buyers' parents operate than to the destinations they do not reach. The strictest version of (30) makes this explicit by adding a firm–year fixed effect, which absorbs the productivity spillover term together with every other destination-neutral determinant of exporting. The access coefficient is then estimated solely from the destination-specific incidence of the linked MNCs' footprints. A pure productivity spillover cannot produce $\zeta_a > 0$ in this design, because it shifts the export entry index equally in all destinations. The comparison of ζ_a with and without the firm–year effect is therefore a specification check: if destination-neutral confounders were loading onto the access treatment, ζ_a would move when they are absorbed. Finally, the time variation in the footprint sets K_{ct} provides a sharper source of identification: when the parent of an already-linked MNC buyer opens an affiliate in a new destination, the access discount switches on in that destination alone, which we exploit in an event-study design around the MNC's expansion.

Identifying productivity spillovers. ζ_θ is identified from the destinations the linked MNCs do *not* reach: there $a_{ict} = 0$ by (29), so an MNC linkage can move the entry index only through the productivity spillover term. This variation is destination-neutral and moves at the firm–year level, so it is only observable when the firm–year component is not absorbed: (30) accordingly includes firm–destination and destination–year fixed effects but no firm–year effects. Identification of ζ_θ then requires that the timing of the first MNC linkage be uncorrelated with other destination-neutral shocks to export participation, conditional on the fixed effects and controls. This is the same identifying assumption that underlies the firm-level analysis, and Section **??** probes it directly, through event-study designs and instrumental variables. ζ_θ is the firm–destination counterpart of the firm-level export-entry effect: both reflect the same structural object, the per-link spillover ψ_{im}, measured on different margins.

Sufficient statistic for the weight of the two mechanisms. Equation (30) delivers a sufficient-statistic decomposition of the total effect of an MNC linkage on export participation. A linkage raises the entry probability by ζ_θ in every destination, and by $\zeta_\theta + \zeta_a$ in the destinations reached by the MNC buyers' parents. Averaged across destinations, the total effect for supplier i is

$$\frac{1}{C}\sum_{c\in\mathcal{C}} \Delta \Pr\left(\text{Export}_{ict}\right) = \zeta_\theta + \zeta_a\, \Omega_{it}, \qquad \Omega_{it} \equiv \frac{1}{C}\left|\{c \in \mathcal{C} : S_{i,t-1} \cap K_{c,t-1} \neq \varnothing\}\right|, \tag{31}$$

where Ω_{it} is the supplier's footprint coverage: the share of destinations in which at least one of its MNC buyers' parents operates. Summing instead of averaging gives the expected number of destinations entered, C times (31); the share in (32) below is identical under either normalization. The productivity channel contributes ζ_θ because it operates in every destination; the market-access channel contributes $\zeta_a \Omega_{it}$ because it operates only where the linked MNCs are present. The share of the total effect attributable to market access is therefore

$$s_a = \frac{\zeta_a \overline{\Omega}}{\zeta_\theta + \zeta_a \overline{\Omega}}, \tag{32}$$

with $\overline{\Omega}$ the average coverage among linked suppliers and $s_\theta = 1 - s_a$ the productivity share. The decomposition is a sufficient statistic in that it requires only the two reduced-form coefficients and the observed coverage, with no additional structural parameters. Note that it is, however, a partial equilibrium decomposition of the weight of both channels: it holds linkage sets, footprints, and prices fixed, allocating the average per-destination entry effect between the two channels rather than computing a counterfactual. The general-equilibrium decomposition, in which link formation and export entry respond jointly, is carried out in the full quantification in Section **??**. We implement (32) in Section **??**.

From estimates to model quantification. Each channel is disciplined by the data that measures it most directly. Access operates only through export entry, so ζ_a disciplines the reductions η_{mc} through the discount (29). The productivity spillovers ψ_{ij}, by contrast, are destination-neutral, so they are disciplined by the firm-level evidence of Section **??**. Because export entry responds nonlinearly to the latent index and the model is static, coefficients from regressions estimated on simulated cross-sections do not recover these structural objects: in the model cross-section, linked suppliers differ systematically in productivity from unlinked ones, so simulated regression coefficients conflate the spillover with selection. The model instead disciplines each channel with a simulated counterfactual contrast that isolates it by construction. For the productivity channel, the contrast is the change in export participation when the productivity spillovers of MNC links are removed, holding the equilibrium fixed; it is matched to the within-firm first-time entry response to a first MNC linkage estimated in Section **??**, both expressed relative to baseline entry. For the access channel, the contrast is the change in entry into covered destinations when the access discounts are set to zero, holding the equilibrium fixed; it is matched to the estimate of ζ_a from the strictest version of (30), with firm–year fixed effects, normalized by baseline entry into uncovered destinations. Each contrast is exactly zero when its channel is shut down, so the two moments inherit the incidence logic of the identification argument above: the productivity contrast is destination-neutral, the access contrast destination-specific. ζ_θ is deliberately left untargeted: the model, disciplined by the firm-level entry response, must reproduce the firm–destination productivity coefficient on its own—an overidentifying check that the two margins agree. The remaining parameters—the elasticities σ, ξ, γ, the Armington pair (ρ, ν), the productivity distribution, the meeting technology $(\lambda_j, \varkappa)$, and the fixed and trade costs—are set by simulated method of moments or taken directly from the data, as detailed in Section **??**.

Testable implications. Beyond the signs of ζ_θ and ζ_a, the structure of the two channels carries sharper predictions, which we take to the data in Sections **??**, **??**, and **??** and in the Online Appendix. The access discount (29) is tied to the linked MNCs' affiliate footprints and operates through export fixed costs, which yields five implications:

1. *Destination concordance.* A supplier linked to an MNC is more likely to export to the destinations its buyer's parent reaches than to those it does not, and the differential survives the absorption of all destination-neutral variation by firm–year fixed effects.

2. *Timing.* The discount switches on when the footprint does: entry into destination c rises only after an already-linked buyer's parent opens an affiliate in c, with no differential trend before.

3. *Third-party buyers.* The discount lowers the cost of serving the destination rather than creating demand from the MNC itself, so entry into covered destinations occurs predominantly through sales to buyers unrelated to the linked MNC.

4. *Backward linkages only.* Access travels through supplying relationships—only MNC buyers enter K_{ct}—so buying from an MNC whose parent is present in c generates no discount and should not predict entry into c.

5. *Extensive margin only.* Because the discount reduces a fixed cost, it operates on the entry margin alone: conditional on exporting to c, footprint overlap should not raise export values.

The productivity spillover, in turn, is destination-neutral and embodied in supplier productivity $\theta_i(S)$, so its fingerprints extend beyond the covered destinations and beyond exporting altogether:

6. *Entry beyond the footprint.* Because the spillover raises productivity in every market, an MNC linkage raises export entry also in destinations the buyer's parent does not reach, where the access discount is zero by construction: $\zeta_\theta > 0$. A pure access story predicts no entry response there.

7. *Firm-level performance.* The gains are visible in outcomes unrelated to the MNC relationship itself: labor productivity, total sales, and in particular sales to other, non-MNC domestic buyers.

8. *Observable upgrading.* The learning behind ψ_{ij} has observable counterparts: upgrading investments after the first MNC sale, and surveyed reports of stricter quality standards, improved production procedures, and better managerial practices.

The same structure yields three additional, more demanding implications. These are useful as specification and heterogeneity tests rather than as the core identifying restrictions:

9. *Heterogeneous learning.* Because $\psi_{ij} = \beta_i^{k(j)}\alpha_j$, productivity and export responses should be larger for suppliers with greater absorptive capacity and for relationships with more productive buyers.

10. *Link complementarity.* Supermodularity implies that, conditional on meeting opportunities, the marginal value of activating another buyer relationship rises with the active set. A payoff-side corollary is that multiple valuable links can generate larger export gains than a single link.

11. *Proximity to the entry threshold.* A productivity improvement or fixed-cost reduction changes export status only for a firm–destination pair sufficiently close to the zero-profit cutoff in (27). Among pairs that have not yet exported, effects should therefore be concentrated among those with higher ex-ante entry propensity.

References

Arkolakis, Costas, Fabian Eckert, and Rowan Shi. “Combinatorial Discrete Choice: Theory and Application to Multinational Production.” *American Economic Review*, Forthcoming.

Jia, Panle. “What Happens When Wal-Mart Comes to Town: An Empirical Analysis of the Discount Retailing Industry.” *Econometrica* 76(6): 1263–1316, 2008.

Topkis, Donald M. *Supermodularity and Complementarity.* Princeton University Press, 1998.

A Appendix

A.1 Supermodularity of the Supplier Objective

This appendix establishes that the supplier objective is supermodular and hence satisfies single-crossing differences in choices from below. The bilateral iceberg cost enters only through the effective demand shifter $\widetilde{D}_{ic} = \tau_{ic}^{1-\sigma} D_c$, a positive firm–destination constant that merely rescales the export term, so every step below holds with D_c read as $\widetilde{D}_{ic}$. The match shifters and the link-cost shifter are likewise positive constants within the supplier's problem, so every step below also holds with B_j read as $m_{ij} B_j$ and f_j read as $\varphi_i f_j$. Fix supplier i and met set M_i. For notational simplicity, suppress the supplier subscript when no confusion arises and write

$$\Phi(S) = \theta_i(S)^{\sigma-1}.$$

The supplier objective is

$$\Pi_i(S) = \frac{\kappa}{\sigma}\Phi(S)\sum_{j\in S} B_j + \sum_{c\in\mathcal{C}}\left[\frac{\kappa}{\sigma}\Phi(S)D_c - w f_c^X(S)\right]^+ - w\sum_{j\in S} f_j, \qquad S \subseteq M_i.$$

Lemma 1 (Productivity-profit complementarity, category-heterogeneous)**.** *Let each buyer j have category $k(j) \in \mathcal{K}$, and let the learning technology be log-additive with category-dependent per-link spillovers,*

$$\log\theta_i(S) = \log\bar{\theta}_i + \sum_{j\in S}\psi_{ij}, \qquad \psi_{ij} = \beta_i^{k(j)}\alpha_j \geq 0.$$

Then, for any $\sigma > 1$, the function $\Phi(S) = \theta_i(S)^{\sigma-1}$ is nondecreasing and supermodular on 2^{M_i}, regardless of how ψ_{ij} varies across buyer categories.

Proof. The productivity technology implies

$$\Phi(S) = \bar{\theta}_i^{\sigma-1}\exp\Big\{(\sigma-1)\sum_{j\in S}\psi_{ij}\Big\}.$$

Because the exponent is additive (modular) in S, for any two buyers $j, j' \notin S$ the cross-difference is

$$D_{jj'}\Phi(S) = \Phi(S)\Big(e^{(\sigma-1)\psi_{ij}} - 1\Big)\Big(e^{(\sigma-1)\psi_{ij'}} - 1\Big) \geq 0,$$

since $\psi_{ij}, \psi_{ij'} \geq 0$ make each factor nonnegative and $\Phi(S) > 0$. Thus Φ is supermodular. Likewise the marginal difference

$$\delta_j\Phi(S) = \Phi(S)\Big(e^{(\sigma-1)\psi_{ij}} - 1\Big) \geq 0,$$

so Φ is nondecreasing. Neither sign depends on equality of the ψ_{ij} across categories. Supermodularity fails only if one of the two maintained properties is dropped: a category with a negative per-link spillover ($\psi_{ij} < 0$) renders a mixed pair submodular, and a non-separable aggregator with diminishing returns across links makes links substitutes rather than complements. □

Lemma 2 (Domestic-profit supermodularity)**.** *Let $R(S) = \sum_{\ell\in S} B_\ell$, with $B_\ell \geq 0$. If Φ is nondecreasing and supermodular, then $S \mapsto \Phi(S)R(S)$ is supermodular.*

Proof. For $j, k \notin S$,

$$D_{jk}\left[\Phi(S)R(S)\right] = R(S)D_{jk}\Phi(S) + B_j\delta_k\Phi(S\cup\{j\}) + B_k\delta_j\Phi(S\cup\{k\}).$$

Each term is nonnegative because $R(S) \geq 0$, $B_j, B_k \geq 0$, and Φ is nondecreasing and supermodular. □

Lemma 3 (Positive-part composition)**.** *Let $I(S)$ be nondecreasing and supermodular. Then $S \mapsto [I(S)]^+$ is supermodular.*

Proof. Fix $j, k \notin S$. Let $a = I(S)$, $b = I(S \cup \{j\})$, $c = I(S \cup \{k\})$, and $d = I(S \cup \{j, k\})$. Since I is nondecreasing, $a \leq b, c \leq d$. Since I is supermodular, $d - a \geq (b - a) + (c - a)$. Let $g(x) = [x]^+$, which is nondecreasing and convex. Since $c \geq a$ and $d - c \geq b - a \geq 0$, the increment of g over $[c, d]$ is a longer step from a weakly higher base point than that over $[a, b]$; monotonicity and convexity of g then give

$$g(d) - g(c) \geq g(b) - g(a),$$

which is equivalent to

$$g(d) + g(a) - g(b) - g(c) \geq 0.$$

Thus $g \circ I$ is supermodular. □

Lemma 4 (Baseline export payoff)**.** *Under additive market-access spillovers,*

$$f_c^X(S) = \bar{f}_c - \sum_{m \in S \cap K_c} \eta_{mc}, \qquad \eta_{mc} \geq 0,$$

the destination-c export payoff

$$S \mapsto \left[\frac{\kappa}{\sigma}\Phi(S)D_c - wf_c^X(S)\right]^+$$

is supermodular.

Proof. Define the inside export payoff

$$I_c(S) = \frac{\kappa}{\sigma}\Phi(S)D_c - wf_c^X(S).$$

Because $D_c > 0$ and Φ is nondecreasing and supermodular by Lemma 1, the first term is nondecreasing and supermodular. Under additive access,

$$-wf_c^X(S) = -w\bar{f}_c + w\sum_{m \in S \cap K_c} \eta_{mc},$$

which is modular and nondecreasing in S (the constant wage $w > 0$ preserves both properties). Hence $I_c(S)$ is nondecreasing and supermodular. Lemma 3 then implies that $[I_c(S)]^+$ is supermodular. □

Proposition 1 (Baseline supermodularity and SCD-C)**.** *Under the baseline productivity and market-access technologies, the supplier objective $\Pi_i(S)$ is supermodular on 2^{M_i}. Consequently, it satisfies single-crossing differences in choices from below.*

Proof. The domestic-profit component is supermodular by Lemma 2. Each export component is supermodular by Lemma 4. The link-cost term $-\sum_{j \in S} f_j$ is modular. Since sums of supermodular and modular functions are supermodular, $\Pi_i(S)$ is supermodular.

Supermodularity is equivalent to increasing marginal values: if $S \subseteq S' \subseteq M_i \setminus \{j\}$, then

$$\delta_j \Pi_i(S') \geq \delta_j \Pi_i(S).$$

Therefore, if $\delta_j \Pi_i(S) \geq 0$, then $\delta_j \Pi_i(S') \geq 0$. This is single-crossing differences in choices from below. □

A.2 Monotone Comparative Statics and Threshold Policies

This appendix records the comparative-statics and policy-structure results that follow from the supermodularity established in Appendix A.1.

Corollary 1 (Monotone comparative statics)**.** *Let $\overline{S}_i$ denote the greatest maximizer of $\Pi_i(S)$ over $S \subseteq M_i$. Under the baseline parameterization, $\overline{S}_i$ is nondecreasing, in the strong set order, in $\bar{\theta}_i$, in each absorptive-capacity slope β_i^k, in each B_j, and in each D_c. It is nonincreasing in each f_j and each $\bar{f}_c$.*

Proof. By Proposition 1, the objective is supermodular in S. It remains to verify increasing or decreasing differences between the choice set and the relevant parameter.

First, $\bar{\theta}_i$ scales $\Phi(S)$ multiplicatively:

$$\Phi(S) = \bar{\theta}_i^{\sigma-1} \exp\left\{(\sigma-1)\sum_{j\in S}\psi_{ij}\right\}.$$

Thus an increase in $\bar{\theta}_i$ raises the return to any active set, and raises it more for sets with higher productivity gains and larger market exposure. The domestic term therefore has increasing differences in $(S, \bar{\theta}_i)$. Each export inside payoff also has increasing differences in $(S, \bar{\theta}_i)$, and the positive-part operator preserves this monotonicity because it is nondecreasing and convex. Hence Π_i has increasing differences in $(S, \bar{\theta}_i)$.

For an absorptive-capacity slope β_i^k, the same expression gives $\delta_j\Phi(S) = \Phi(S)\big(e^{(\sigma-1)\psi_{ij}} - 1\big)$; for a link to a category-k buyer this is increasing in β_i^k because both $\Phi(S)$ and the bracketed term rise. The marginal value of every category-k link is therefore larger at higher β_i^k, so Π_i has increasing differences in (S, β_i^k) for each k.

Second, for a buyer demand shifter B_h,

$$\frac{\partial}{\partial B_h}\left[\frac{\kappa}{\sigma}\Phi(S)\sum_{j\in S}B_j\right] = \frac{\kappa}{\sigma}\Phi(S)\mathbf{1}\{h\in S\},$$

which is nondecreasing in S. Hence Π_i has increasing differences in (S, B_h).

Third, for a destination demand shifter D_c, the only affected term is

$$\left[\frac{\kappa}{\sigma}\Phi(S)D_c - wf_c^X(S)\right]^+.$$

Because $\Phi(S)$ is nondecreasing in S, the marginal effect of D_c is weakly larger for larger active sets. Hence Π_i has increasing differences in (S, D_c).

Fourth, for a link cost f_h,

$$\frac{\partial}{\partial f_h}\left[-w\sum_{j\in S}f_j\right] = -w\mathbf{1}\{h\in S\},$$

which is nonincreasing in S. Hence Π_i has decreasing differences in (S, f_h).

Finally, for a baseline export fixed cost $\bar{f}_c$, the affected term is

$$\left[\frac{\kappa}{\sigma}\Phi(S)D_c - w\bar{f}_c + w\sum_{m\in S\cap K_c}\eta_{mc}\right]^+.$$

The marginal effect of increasing $\bar{f}_c$ is weakly more negative for larger active sets, because larger active sets are more likely to make the destination profitable. Hence Π_i has decreasing differences in $(S, \bar{f}_c)$.

Topkis's monotone comparative statics theorem then implies that the greatest maximizer is nondecreasing in parameters with increasing differences and nonincreasing in parameters with decreasing differences. □

Assumption 1 (Generic no ties)**.** For almost every value of supplier productivity $\bar{\theta}_i$, the supplier problem has a unique maximizer. Equivalently, for any two distinct feasible link sets $S \neq S'$, the equality $\Pi_i(S) = \Pi_i(S')$ holds only on a set of $\bar{\theta}_i$ values with measure zero.

Corollary 2 (Threshold policy in supplier productivity)**.** *Under Assumption 1, the optimal link policy is a step function of $\bar{\theta}_i$, with changes only at finitely many indifference thresholds.*

Proof. The feasible set 2^{M_i} is finite. Collect the export-entry breakpoints in $z = \bar{\theta}_i^{\sigma-1}$ across all candidate link sets $S \subseteq M_i$ and destinations c; these form a finite partition of z. On each cell every candidate S has a fixed export set, so the objective takes the affine form $\Pi_i(S;\bar{\theta}) = z\,G(S) - K(S)$ for constants depending on that fixed export set, and the payoff difference $\Pi_i(S) - \Pi_i(S')$ is affine in z. Hence each pairwise indifference equation has finitely many solutions in z, except in the nongeneric case in which the two payoffs coincide on an interval. Assumption 1 rules out such nongeneric intervals. Taking the union over the finite number of pairs S, S' yields finitely many indifference thresholds. Between any two adjacent thresholds, the unique maximizer is constant. Therefore the optimal link policy is a step function of $\bar{\theta}_i$. □

A.3 General Equilibrium: Aggregation, the Operator, and Existence

This appendix provides the aggregation formulas referenced in Section 1.3, constructs the equilibrium operator, and proves existence.

Exact aggregation. Each supplier linked to buyer j contributes $m_{ij}\, p_i^{1-\sigma} = \kappa\, m_{ij}\, \theta_i(S_i)^{\sigma-1}$ to the price index (3). Decomposing realized productivity as $\theta_i(S)^{\sigma-1} = \bar{\theta}_i^{\sigma-1}\widetilde{\Phi}_i(S)$ with $\widetilde{\Phi}_i(S) = \exp\{(\sigma-1)\sum_{j\in S}\psi_{ij}\}$, the price index aggregates supplier policies as

$$P_j^{1-\sigma} = \kappa \int m_{ij}\, \bar{\theta}_i^{\sigma-1}\, \widetilde{\Phi}_i(S_i^\star)\, \mathbf{1}\{j \in S_i^\star\}\, d\mu(\bar{\theta}_i, \beta_i, \varphi_i, m_i, M_i),$$

where $m_i = (m_{ij})_{j\in\mathcal{J}}$, μ is the joint distribution of supplier types, match shifters, and met sets, and $S_i^\star = S^\star(\bar{\theta}_i, \beta_i, \varphi_i, m_i, M_i; B)$ is the optimal policy. With a continuum of suppliers, an exact law of large numbers makes this aggregate deterministic.

Incomplete moments. By Corollary 2, for a fixed met set M, absorptive-capacity type β, cost shifter φ, and match shifters the optimal policy is a step function of $\bar{\theta}$, constant at $S^{(k)}$ on intervals (e_{k-1}, e_k) with $0 = e_0 < e_1 < \cdots < e_K = \infty$. The conditional aggregate then reduces to a finite sum of incomplete moments,

$$\int \bar{\theta}^{\sigma-1}\widetilde{\Phi}(S^\star)\, \mathbf{1}\{j \in S^\star\}\, dF(\bar{\theta}) = \sum_{k=1}^{K} \widetilde{\Phi}(S^{(k)})\, \mathbf{1}\{j \in S^{(k)}\}\, \mathcal{I}(e_{k-1}, e_k), \qquad \mathcal{I}(a,b) \equiv \int_a^b \bar{\theta}^{\sigma-1}\, dF(\bar{\theta}).$$

On any range of $\bar{\theta}$ over which the optimal export set X is fixed, the objective is $\bar{\theta}$-separable, since $\theta_i(S)^{\sigma-1} = \bar{\theta}^{\sigma-1}\widetilde{\Phi}(S)$:

$$\Pi_i(S;\bar{\theta}) = \bar{\theta}^{\sigma-1} G_X(S) - K_X(S),$$
$$G_X(S) = \frac{\kappa}{\sigma}\widetilde{\Phi}(S)\Big(\sum_{j\in S} m_{ij}B_j + \sum_{c\in X}\widetilde{D}_{ic}\Big), \qquad K_X(S) = w\Big(\sum_{c\in X} f_c^X(S) + \varphi_i \sum_{j\in S} f_j\Big).$$

The export set X is itself nondecreasing in $\bar{\theta}$, with a finite number of breakpoints (each destination enters once its export payoff turns positive), so for fixed S the objective is convex and piecewise of this form. The breakpoints of the optimal policy are accordingly the union of two finite families: the *export-entry* thresholds, at which the optimal export set of a candidate S gains a destination, and the *link-set* thresholds, at which the optimal active set changes. The latter solve the indifference conditions $\Pi_i(S^{(k-1)};\bar{\theta}) = \Pi_i(S^{(k)};\bar{\theta})$ and are closed form once we use the *branch-specific* export sets $X^{(k-1)}, X^{(k)}$ that are optimal for $S^{(k-1)}, S^{(k)}$ on either side of the threshold,

$$e_k = \left[\frac{K_{X^{(k)}}(S^{(k)}) - K_{X^{(k-1)}}(S^{(k-1)})}{G_{X^{(k)}}(S^{(k)}) - G_{X^{(k-1)}}(S^{(k-1)})}\right]^{1/(\sigma-1)}.$$

The common-X form is the special case in which both candidate sets share the same optimal export set on the relevant range. The incomplete moment $\mathcal{I}(a,b)$ is closed form for the productivity distributions we use. Writing $t = \sigma - 1$: for Pareto F with lower bound $\bar{\theta}_{\min}$ and shape $k > t$, and $a' = \max\{a, \bar{\theta}_{\min}\}$,

$$\mathcal{I}(a,b) = \frac{k\,\bar{\theta}_{\min}^{k}}{t-k}\left[b^{\,t-k} - a'^{\,t-k}\right] \qquad (b > \bar{\theta}_{\min});$$

for log-normal $\log\bar{\theta} \sim N(\mu, s^2)$,

$$\mathcal{I}(a,b) = e^{t\mu + t^2 s^2/2}\left[\Phi_N\Big(\tfrac{\ln b - \mu - ts^2}{s}\Big) - \Phi_N\Big(\tfrac{\ln a - \mu - ts^2}{s}\Big)\right],$$

with Φ_N the standard normal cdf.

The buyer block and business stealing. Given the price indices $\{P_j\}$, the buyer block returns the demand shifters $B_j = P_j^{\sigma-1}E_j$ and the elasticity (23). Cost minimization over the nest (2) gives the unit cost of the composite–labor bundle, $c_j = [\omega^\xi P_j^{1-\xi} + (1-\omega)^\xi w^{1-\xi}]^{1/(1-\xi)}$, and the intermediate cost share $s_j^M = \omega^\xi P_j^{1-\xi}/c_j^{1-\xi}$, so by Shephard's lemma $\partial \ln c_j/\partial \ln P_j = s_j^M$. The buyer prices at the constant markup over c_j/α_j and faces final demand of elasticity γ, so its revenue is $R_j^Y \propto c_j^{1-\gamma}$ and its expenditure on the composite is $E_j = s_j^M \frac{\gamma-1}{\gamma} R_j^Y$. Hence $\ln B_j = (\sigma-1)\ln P_j + \ln s_j^M + \ln R_j^Y + \text{const}$. Using $\partial \ln s_j^M/\partial \ln P_j = (1-\xi)(1-s_j^M)$ and $\partial \ln R_j^Y/\partial \ln P_j = (1-\gamma)s_j^M$,

$$\frac{\partial \ln B_j}{\partial \ln P_j} = (\sigma-1) + (1-\xi)(1-s_j^M) + (1-\gamma)s_j^M = (\sigma-\xi) + (\xi-\gamma)s_j^M,$$

which is (23). It is affine in $s_j^M \in [0,1]$, with value $\sigma - \xi$ at $s_j^M = 0$ and $\sigma - \gamma$ at $s_j^M = 1$; hence it is positive for all buyers if and only if $\sigma > \xi$ and $\sigma > \gamma$, i.e. $\sigma > \max\{\xi, \gamma\}$. Under that condition a fall in P_j lowers B_j: business stealing among a buyer's suppliers.

The equilibrium operator. Collect the demand shifters $B = (B_j)_{j \in \mathcal{J}}$ in a compact box $\mathcal{B} \subset \mathbb{R}^J_{++}$. Define $\Psi : \mathcal{B} \to \mathcal{B}$ by: given B, each supplier solves its problem (Section 1.2); aggregate the resulting supplier prices into the price indices $\{P_j(B)\}$ by the exact-aggregation step above; apply the buyer block to obtain new shifters

$$\Psi_j(B) = P_j(B)^{\sigma-1} E_j(B).$$

An equilibrium is a fixed point $B = \Psi(B)$.

Proposition 2 (Existence). *Ψ is a continuous self-map of the nonempty compact convex set $\mathcal{B}$ and therefore has a fixed point.*

Proof. We take $\mathcal{B}$ to be a box $[\underline{B}, \overline{B}] \subset \mathbb{R}^J_{++}$ on which every buyer is served by a positive mass of suppliers, so that each $P_j(B)$ is finite, and which Ψ maps into itself.[5] Under the generic no-ties condition (Assumption 1), the optimal policy changes on a μ-null set of types as B varies, so the integrand in the price-index aggregate is continuous in B for μ-almost every type and is dominated by the integrable envelope $\kappa\, m_{ij}\, \bar{\theta}_i^{\sigma-1}$, which is μ-integrable by the finite-moment assumption together with the independence and unit mean of the match shifters; dominated convergence gives continuity of $B \mapsto \{P_j(B)\}$, and the buyer block is continuous, so Ψ is continuous. As a continuous self-map of a nonempty compact convex set, Ψ has a fixed point by Brouwer's theorem. □

Business stealing and multiplicity. Under $\sigma > \max\{\xi, \gamma\}$ the demand-shifter elasticity (23) is positive, embedding a force of strategic substitutability: a larger B raises supplier participation, lowers the price indices, and, through (23), feeds back into lower shifters. This countervailing feedback can make Ψ *antitone* (order-reversing) rather than a contraction, so we do not rule out multiple equilibria a priori. When Ψ is antitone, Ψ^2 is isotone; iterating Ψ^2 from a low and a high shifter bracket, on a region where every buyer is served, then brackets the fixed points of Ψ within that region. We compute these brackets at the aggregate scale Θ implied by the household budget at the labor-clearing wage: collapse of the bracket to a single point that also satisfies $\Psi(B) = B$ certifies a unique equilibrium at that Θ and within the served bracket, whereas a bracket that stays open with a nonzero Ψ-residual reflects a period-two orbit or genuine multiplicity. This diagnostic concerns the demand-shifter map at fixed Θ; a global statement about the joint (B, Θ, w) equilibrium would additionally require uniqueness of the budget-determined $\Theta(B)$ and of the labor-clearing wage. Proposition 3 gives a sufficient condition under which the question does not arise.

Proposition 3 (Uniqueness). *If Ψ is a contraction in the sup-norm of log shifters, the equilibrium is unique.*

Proof. A self-map of a complete metric space with Lipschitz modulus below one has a unique fixed point by the Banach fixed-point theorem. □

Economically, the contraction condition is met when the buyer-demand and supplier-participation responses are not too strong, so that the business-stealing feedback does not compound into self-reinforcing multiplicity.

[5] Such an invariant box exists under two maintained conditions. First, a *positive-service* condition: the meeting technology and the support of supplier types guarantee that, for every $B \in \mathcal{B}$, a positive mass of suppliers activates a link to each buyer j, so $P_j(B) < \infty$. Second, the productivity distribution has a *finite* $(\sigma - 1)$*-th moment* (e.g. Pareto with shape $k > \sigma - 1$, or log-normal), which bounds the price-index aggregate above and, with positive service, below; these bounds furnish $\underline{B}, \overline{B}$ and make $\mathcal{B}$ invariant under Ψ.